

4 (a) amplitude = 1.8 cm *and* period = 0.30 s A1 [1]

(b) $E_K = \frac{1}{2}m \omega^2 (x_0^2 - x^2)$ or $E_K = \frac{1}{2}mv^2$ and $v = \pm \omega \sqrt{(x_0^2 - x^2)}$ C1
 $= \frac{1}{2} \times 0.080 \times (2\pi/0.30)^2 \times [(1.8 \times 10^{-2})^2 - (1.2 \times 10^{-2})^2]$ C1
 $= 3.2 \times 10^{-3} \text{ J}$ A1 [3]